

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Statement of the main problem

- We are now interested in solving the following coupled ode system

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases} \quad (1)$$

- So we want our drivers to solve the following general case as efficiently as possible

$$i \partial_t \phi_\omega(x, t) = H(t) \phi_\omega(x, t) + e^{i\omega t} S(t), \quad H(t) = H_0 + F(t)x, \quad (2)$$

- Considering, the solution to the homogeneous problem, without the source

$$i \partial_t U(t, t_0) = (H_0 + F(t)x) U(t, t_0), \quad U(t, t) = \mathbb{1}, \quad (3)$$

we can then, from the Duhamel principle, write

$$\phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t_0) - i \int_{t_0}^t U(t - \tau) e^{i\omega \tau} S(\tau) d\tau. \quad (4)$$

The above is the exact formulation of our problem.

Geometric propagators

Norm-preserving propagation schemes

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U(t + \Delta t - \tau) e^{i\omega\tau} S(\tau) d\tau. \quad (4)$$

- Our test system will be a stationary QHO, with source, i.e

$$H_a = -\frac{1}{2}\partial_x^2 + \frac{1}{2}x^2, \quad F(t) = 0$$

$$\begin{cases} i\partial_t \psi(x, t) &= \left(\hat{H}_a + F(t)x \right) \psi(x, t), & (\hbar = \omega = m = 1) \\ i\partial_t \phi_\omega(x, t) &= \left(\hat{H}_a + F(t)x \right) \phi_\omega(x, t) - ie^{i\omega t} \partial_x \psi(x, t), \end{cases} \quad (5)$$

- The goal is to develop a global order 4 scheme, even for the non trivial forced QHO case $F(t) \neq 0$
- The roadmap is the following
 - Midpoint Duhamel (order 2)
 - Yoshida symmetric decomposition (order 4)
 - Simpson quadrature (order 4)

Geometric propagators

Convergence criteria

$$\partial_t z = -iHz - ie^{i\omega t} S(y, t). \quad (6)$$

- The scheme are assessed based on 2 different tests with the number of steps N , and $T = t_f - t_i$, such as

$$N_\ell = N_0 2^\ell, \quad h_\ell = \frac{T}{N_0 2^\ell}. \quad (7)$$

- Absolute convergence test (applicable only when $F(t) = 0$)

$$p_\ell^{\text{abs}} = \frac{\log(E_{\ell-1}^{\text{abs}}/E_\ell^{\text{abs}})}{\log(2)}, \quad \text{with} \quad E_\ell^{\text{abs}} = |z_\ell(t_f) - z_{\text{exact}}(t_f)|. \quad (8)$$

- Self-consistency test (Richardson's)

$$p_\ell^{\text{self}} = \frac{\log(D_{\ell-1}/D_\ell)}{\log(2)} \quad \text{with} \quad D_\ell = |z_\ell(t_f) - z_{\ell-1}(t_f)|, \quad D_\ell \sim Ch_\ell^p. \quad (9)$$

- The absolute convergence test tells reveals potential error in the model

- The Richardson's self-consistency test tells you if your scheme works or not



- We construct the solution by setting $S(x, t) = \psi_0(x)[1]$, $\psi(x, t)[2]$, then $\hat{p}\psi(x, t)[3]$

Geometric propagators

Midpoint Duhamel - 2nd order approximation

$$\phi_\omega(x, t + \Delta t) = U(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U(t + \Delta t - \tau) e^{i\omega\tau} S(\tau) d\tau. \quad (4)$$

- We can show that the above can be expressed as

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - e^{i\omega t_{n+1/2}} \partial_x \psi_{n+1/2}, \quad (10)$$

where the source is evaluated at $t_{n+1/2} = t + \Delta t/2$, i.e and $\psi_{n+1/2} = U(\Delta t/2)\psi_n$.

- This is the midpoint Duhamel formula, of order 2.
- It's what will allow us to get familiar with the numerical intricacies of our problem
- The driver shall be regarded as decent if the expected order is met

Geometric propagators

Yoshida (1990) - 4th-order symmetric decomposition

$$\phi_\omega(x, t + \Delta t) = U_4(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau) e^{i\omega\tau} S(\tau) d\tau. \quad (4)$$

- From the following Strang operator $U(h) = e^{-iE_n h/2} e^{-iV_{\text{ext}}(t+h/2)h} e^{-iE_n h/2}$, we compose

$$U_4(h) = U(c_1 h) U(c_2 h) U(c_1 h), \quad \text{with} \quad c_1 = \frac{1}{2 - 2^{1/3}}, \quad c_2 = -\frac{2^{1/3}}{2 - 2^{1/3}}. \quad (11)$$

- More precisely, we can observe the following

Theorem 8.6 (Hairer, Solving ODE II p.554)

If Ψ_h represent a one-step method that is symmetric, of order $p = 2k$ and defined in the domain, then the relations

$$2c_1 + c_2 = 1, \quad 2c_1^{p+1} + c_2^{p+1} = 0, \quad (12)$$

imply that the composition (11) is of order $p + 2$.

If you start from the Strang method, then in the above $k = 1$.

Geometric propagators

Simpson Duhamel - 4th-order quadrature

$$\phi_\omega(x, t + \Delta t) = U_4(t + \Delta t, t)\phi_\omega(x, t) - i \int_t^{t+\Delta t} U_4(t + \Delta t - \tau) e^{i\omega\tau} S(\tau) d\tau. \quad (4)$$

- The simpson quadrature consists in writing

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (13)$$

where the source is evaluated at t_n , $t_{n+\frac{1}{2}}$ and t_{n+1}

- This is the simpson Duhamel formula, of order 4.
- It requires a 4th order evolution operator to work
- The driver shall be regarded as decent if the expected order is met

Geometric propagators

Simpson Duhamel - Absolute convergence test results - stationary case [2]

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \tag{13}$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \tag{11}$$

t	$\Re \phi(1)_{\text{num}}$	$\Im \phi(1)_{\text{num}}$	$\ \phi_{\text{num}}\ $	$\Re \phi(1)_{\text{ex}}$	$\Im \phi(1)_{\text{ex}}$	$\ \phi_{\text{ex}}\ $
-1.0000	1.000000	0.000000	1.000000	1.000000	0.000000	1.000000
-0.7500	0.938024	-0.364441	1.006333	0.955750	-0.292527	1.000000
-0.5000	0.810332	-0.712678	1.079142	0.810332	-0.712678	1.079142
-0.2500	0.620011	-1.034887	1.206402	0.662892	-0.972983	1.178087
0.0000	0.371494	-1.321754	1.372968	0.371494	-1.321754	1.372968
0.2500	0.070478	-1.564653	1.566239	0.134567	-1.519977	1.525923
0.5000	-0.276172	-1.755810	1.777397	-0.276172	-1.755810	1.777397
0.7500	-0.660547	-1.888450	2.000641	-0.581066	-1.866891	1.955140
1.0000	-1.073833	-1.956933	2.232197	-1.073833	-1.956933	2.232197

$$\text{Ne} = 2, \quad \text{Np} = 6$$

$$-\text{xmin} = \text{xrange}/2 = 7$$

$$t_1 = -t_0 = T/2 = 1$$

Initial conditions and parameters

$$\psi_0(x) = (\pi)^{-1/4} e^{-(x-x_c)^2/2}$$

$$\phi_0^\omega(x) = \psi_0(x)$$

$$S(x,t) = \psi(x,t)$$

$$F(t) = F_0 \cos(\omega t)$$

$$x_c = F_0 = 0, \quad \omega = 1.$$

Level	Δt (relative)	E_ψ	E_ϕ	Order (p_ψ/p_ϕ)
1	Δt_0	1.656925(-14)	1.173076(-11)	—
2	$\Delta t_0/2$	3.154099(-14)	7.791598(-13)	-0.93 / 3.91
3	$\Delta t_0/4$	5.027708(-14)	1.416639(-13)	-0.67 / 2.46
4	$\Delta t_0/8$	1.070034(-13)	2.478027(-13)	-1.09 / -0.81
Observed order		-1.0897 / -0.8067		

■ Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

Geometric propagators

Simpson Duhamel - Absolute convergence test results - stationary case [3]

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (13)$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad (11)$$

t	$\Re \phi_{\text{num}}$	$\Im \phi_{\text{num}}$	$\ \phi_{\text{num}}\ $	$\Re \phi_{\text{ex}}$	$\Im \phi_{\text{ex}}$	$\ \phi_{\text{ex}}\ $
-1.0000	0.000000	0.707107	0.707107	0.000000	0.000000	0.000000
-0.7500	0.387890	0.616635	0.728490	0.372890	0.725260	0.815604
-0.5000	0.722462	0.394582	0.823193	0.914186	0.549921	1.066741
-0.2500	0.963989	0.064716	0.966159	1.351538	0.153071	1.360186
0.0000	1.082191	-0.337874	1.133709	1.545652	-0.208745	1.559693
0.2500	1.059302	-0.770291	1.309759	1.640943	-0.622311	1.755448
0.5000	0.891893	-1.186125	1.484037	1.576669	-1.280872	2.033065
0.7500	0.591262	-1.540110	1.649705	1.392728	-1.703561	2.198913
1.0000	0.182351	-1.792656	1.801907	0.922767	-2.243259	2.427810

Level	Δt (relative)	E_ψ	E_ϕ	Order (p_ψ/p_ϕ)
1	Δt_0	1.656925(-14)	8.667525(-1)	—
2	$\Delta t_0/2$	3.154099(-14)	8.667525(-1)	-0.93 / 0.00
3	$\Delta t_0/4$	5.027708(-14)	8.667525(-1)	-0.67 / 0.00
4	$\Delta t_0/8$	1.070034(-13)	8.667525(-1)	-1.09 / 0.00
Observed order		-1.0897/0.0000		

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$$\text{Ne} = 2, \quad \text{Np} = 6$$

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Initial conditions and parameters

$$\psi_0(x) = (\pi)^{-1/4} e^{-(x-x_c)^2/2}$$

$$\phi_0^\omega(x) = -i\partial_x \psi_0(x)$$

$$S(x, t) = -i\partial_x \psi(x, t)$$

$$F(t) = F_0 \cos(\omega t)$$

$$x_c = F_0 = 0, \quad \omega = 1.$$

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$

Geometric propagators

Simpson Duhamel - Richardon's self-consistency test results - stationary case [3]

$$\phi_{n+1}^\omega = U_4(\Delta t)\phi_n^\omega - \frac{i\Delta t}{6} \left(U_4(\Delta t)e^{i\omega t_n} S_n + 4U_4(\Delta t/2)e^{i\omega t_{n+\frac{1}{2}}} S_{n+\frac{1}{2}} + U_4(0)e^{i\omega t_{n+1}} S_{n+1} \right) \quad (13)$$

$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad (11)$$

t	$\Re \psi(1)$	$\Im \psi(1)$	$\ \psi\ $	$\Re \phi(2)$	$\Im \phi(2)$	$\ \phi\ $
-1.0000	1.000000	0.000000	1.000000	0.000000	0.707107	0.707107
-0.7500	0.992713	-0.120505	1.000000	0.387890	0.616635	0.728490
-0.5000	0.970957	-0.239253	1.000000	0.722462	0.394582	0.823193
-0.2500	0.935051	-0.354514	1.000000	0.963989	0.064716	0.966159
0.0000	0.885516	-0.464608	1.000000	1.082191	-0.337874	1.133709
0.2500	0.823076	-0.567931	1.000000	1.059302	-0.770291	1.309759
0.5000	0.748640	-0.662977	1.000000	0.891893	-1.186125	1.484037
0.7500	0.663292	-0.748360	1.000000	0.591262	-1.540110	1.649705
1.0000	0.568278	-0.822837	1.000000	0.182351	-1.792656	1.801907

Level	Δt (relative)	E_ψ	E_ϕ	Order (p_ψ/p_ϕ)
1	Δt_0	1.513885(-14)	2.613033(-08)	—
2	$\Delta t_0/2$	2.033966(-14)	1.632841(-09)	— / 4.0000
3	$\Delta t_0/4$	5.684045(-14)	1.019863(-10)	— / 4.0009
Observed order				— 1.4826/4.0009

- Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

$$N_e = 2, \quad N_p = 6$$

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Simpson Duhamel - Richardon's self-consistency test results - nonstationary case [3]

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$$U_4(t) = U(c_1 h)U(c_2 h)U(c_1 h), \quad (11)$$

t	$\Re \psi(1)$	$\Im \psi(1)$	$\ \psi\ $	$\Re \phi(2)$	$\Im \phi(2)$	$\ \phi\ $
-1.0000	1.000000	0.000000	1.000000	0.000000	0.707107	0.707107
-0.7500	0.985794	-0.119190	1.000000	0.385146	0.609970	0.728679
-0.5000	0.937711	-0.226288	1.000000	0.696725	0.368848	0.825288
-0.2500	0.854960	-0.305377	1.000000	0.876025	0.027341	0.972590
0.0000	0.748376	-0.343661	1.000000	0.896498	-0.345598	1.143971
0.2500	0.637355	-0.338371	1.000000	0.770350	-0.684150	1.320469
0.5000	0.542795	-0.299476	1.000000	0.532603	-0.944268	1.492459
0.7500	0.478888	-0.247266	1.000000	0.223117	-1.102731	1.653612
1.0000	0.447387	-0.205441	1.000000	-0.119191	-1.147779	1.796201

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$$x_c = F_0 = 0, \quad \omega = 1.$$

Level	Δt (relative)	E_ψ	E_ϕ	Order (p_ψ/p_ϕ)
1	Δt_0	5.506211(-06)	1.311796(-05)	—
2	$\Delta t_0/2$	3.439466(-07)	8.271523(-07)	4.000 / 3.988
3	$\Delta t_0/4$	2.150284(-08)	5.193475(-08)	3.9996 / 3.9934
Observed order				3.9996/3.9934

■ Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$